

TileLoc: Sequential Visual Place Recognition of UAVs using Public Tile Maps

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Abstract—We propose *TileLoc*, a visual place recognition method for uncrewed aerial vehicles (UAVs) that uses publicly available web tile maps in GNSS-degraded or GNSS-denied environments without prior mapping. Visual place recognition with web tile maps is challenging because the query image and the map tiles often differ in scale, rotation, and coverage. As a result, single-frame retrieval can be unreliable. Based on our preliminary experiments, we identified major failure cases and designed a simple yet effective pipeline to mitigate them. *TileLoc* builds a multi-zoom, overlapped tile database, uses a set of rotated query views, and stabilizes retrieval results by accumulating evidence over time using log-odds voting with temporal decay. After the initial global search, *TileLoc* switches to local search when its confidence becomes high enough, reducing confusion caused by visually similar tiles in global search. Experimental results showed consistent recall improvements over a strong global retrieval baseline, achieving up to 84.5% Recall@5 within a 250 m threshold. These results demonstrate the effectiveness of *TileLoc* for UAV visual place recognition using public tile maps in GNSS-degraded or GNSS-denied environments.

Index Terms—Visual place recognition, sequential matching, UAV, web tile map.

I. INTRODUCTION

UAVs are increasingly used in disaster monitoring, infrastructure inspection, and defense operations. These operations require reliable navigation over wide areas. However, localization systems must also account for the cost of building and maintaining large-scale reference maps.

GNSS (global navigation satellite system) provides global positioning and is widely used for uncrewed aerial vehicle (UAV)

navigation. However, GNSS accuracy can degrade in GNSS-degraded environments, such as urban canyons, mountainous areas, and dense foliage, due to signal blockage and multipath interference [1]. GNSS is also vulnerable to intentional radio-frequency interference (jamming and spoofing). These limitations motivate visual localization as a complementary solution. Cameras are lightweight, low-power, and inexpensive. They also provide rich environmental cues that can support localization when GNSS performance degrades [2], [3].

A standard vision-based solution in GNSS-denied settings is simultaneous localization and mapping (SLAM), which estimates pose while constructing a map online. SLAM has demonstrated strong performance for navigation [4], [5], [6], [7]. However, wide-area aerial SLAM introduces practical burdens, including prior exploration, map construction, long-term maintenance, and drift management over long trajectories. Moreover, as the map size increases, tracking and optimizing a large number of features requires significant computation and makes real-time operation difficult [8].

In this work, we propose *TileLoc*¹, an aerial visual place recognition (VPR) method that leverages publicly available tile maps as a reference. Tile maps provide broad coverage and immediate accessibility, which reduces the overhead of building and maintaining a dedicated prior map. However, VPR with tile maps is challenging. UAV queries and satellite tiles exhibit differences in viewpoint, illumination, and time-varying appearance. Moreover, UAV images often overlap with only a small region of a fixed field-of-view (FoV) tile. In addition, reference imagery is available only at discrete zoom levels, which can cause scale mismatch across varying UAV altitudes. These challenges were quantitatively analyzed in our preliminary experiments in Section III.

TileLoc addresses these challenges with four pipeline modules. First, it constructs a multi-zoom, overlapped tile database to reduce coverage and scale mismatches. Second, from a single query image, it generates a rotated query batch to improve the rotation robustness of global descriptors. Third, adaptive cumulative-mass candidate selection adjusts the number of retrieved candidates according to the ambiguity of the similarity distribution. Fourth, sequential log-odds evidence aggregation with local-search mode switching uses continuous UAV trajectories to stabilize retrieval. It aggregates retrieval evidence in

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¹The name, *TileLoc*, was inspired by our baseline, *AnyLoc* [9], reflecting our use of web tile maps.

a tile-quantized space and enables reliable switching between global and local image retrieval, reducing confusion caused by visually similar tiles in global search. Section IV describes these four modules. In addition, Section V presents experimental results, and Section VI discusses conclusions, limitations, and future research directions.

II. RELATED WORK

Visual place recognition (VPR) has been widely used for visual localization and SLAM by retrieving the most similar reference images from a database. UAV localization with tile maps is closely related to cross-view geo-localization [10]. Its main difficulties arise from cross-view heterogeneity and limited overlap between query and reference imagery. Existing VPR methods can be categorized into three groups: local feature-based approaches, global feature-based approaches, and sequential approaches.

A. Local Feature-Based Approaches for VPR

Local feature [11], [12], [13], [14], [15] approaches match local keypoints combined with geometric verification or voting. These methods are robust to partial overlap, rotation, and scale variation. However, they often struggle in cross-view settings that involve significant appearance changes due to weather conditions and seasons. To tackle such severe mismatches using local features, Sanlan et al. [16] employed CycleGAN for visual domain adaptation, particularly for filtering-based localization.

Furthermore, exhaustive matching of local features is computationally expensive, and their geometric verification further increases computational cost, which makes wide-area search difficult to run in real time. Several works tackle the computational cost of local feature matching by constraining the search space before expensive geometric verification. A common strategy is to assume a coarse prior and retrieve only local satellite candidates, then refine the match with keypoint descriptors and geometric verification with RANSAC [17]. Other methods divide the satellite map into tile blocks and apply a coarse-to-fine search, where robust matching is applied only to a small set of high-scoring blocks [18]. These designs reduce unnecessary comparisons, but their computational cost remains problematic in large-scale settings.

B. Global Feature-Based Approaches for VPR

Global feature approaches for VPR represent reference images with a compact embedding (e.g., holistic descriptor) [19], enabling efficient large-scale retrieval. Several pipelines further improved robustness by combining fast global retrieval using NetVLAD [20] with a lightweight verification stage, where top-ranked candidates were refined using local feature matching [21]. AnyLoc [9] used transformer-based foundation model features as dense descriptors and aggregated them into global embeddings. Revisit Anything [22] moved beyond a single holistic descriptor by semantic segmentation and segment-wise retrieval, which reduced the impact of clutter and partial mismatch. Recently, MM-Geo [23] addressed multi-scale and multi-positive

tile issues by fine-tuning DINOv2 features with contrastive learning and patch-level reranking. While this patch-wise approach partially mitigates these two issues, it does not address the limited rotational invariance of DINOv2 features and does not exploit the advantages of sequential matching.

C. Sequential Approaches for VPR

To improve the accuracy and reliability of single-frame retrieval, sequential VPR utilizes temporal consistency by matching sequences of images instead of isolated frames [24], [25]. Early approaches such as SeqSLAM [26] searched for consistent matches over a short time window. They were robust to severe appearance changes. Later work treated sequence matching as a probabilistic problem. For instance, some methods used hidden Markov models (HMM) with Viterbi decoding to handle variable speeds and data discontinuities [27].

More recent methods learn sequence-aware representations using deep learning. SeqNet [28] and SeqVLAD [29] condensed a short sequence into a compact descriptor. SeqMatchNet [30] integrated sequence matching with contrastive learning, optimizing descriptors directly for consistency across the sequence. This enabled efficient retrieval without the need for computationally expensive frame-by-frame matching. Multi-sequence maps further improved robustness by using data from multiple trips through the same environment [31].

However, many sequential pipelines rely on reference data that is ordered by time along a path. This structure does not fit well with satellite tile maps, which are organized by space rather than time. Yang et al. [32] tracked position belief through Bayesian updates. They also highlighted practical issues such as featureless regions (e.g., water).

Our approach: In contrast, TileLoc targets a practical setting where reference data are indexed by space (tiles), not by time-ordered traversals. We therefore combine scalable global retrieval with a lightweight voting mechanism for sequential matching in tile space. This design can preserve real-time efficiency without the need for model fine-tuning or complex continuous state estimation, while improving robustness to coverage variation and large in-plane rotations through a multi-zoom overlapped database and rotated query batch.

III. PRELIMINARY ANALYSIS

We conducted a series of preliminary experiments for the following purposes. First, we compared local feature and global feature approaches for VPR on a high-altitude VPR dataset. Second, we analyzed the robustness of global descriptors by applying synthetic transformations (e.g., coverage, rotation, and aspect ratio variations) to simulate challenges in UAV-to-tile map matching scenarios.

Motivation: These experiments were motivated by known properties of global descriptors based on DINOv2 [33] features. DINOv2 training included near-duplicate filtering via SSCD (self-supervised copy detection) [34] and employed simple augmentations such as random resized crops and flips. As a result, robustness to large in-plane rotations may have been limited.

TABLE I
COMPARISON OF LOCAL AND GLOBAL FEATURES FOR VPR ON THE VPAIR DATASET

	Method	Geo ⁽²⁾	Recall (%)			Processing Time (s)
			Top-1	Top-2	Top-5	
Local	ELoFTR [13]	–	89.5	90.0	90.5	12.66
		✓	89.5	90.0	90.5	13.94
	ELoFTR	–	96.5	97.0	97.5	13.06
	(MatchAnything [15])	✓	80.7	85.8	87.8	13.52
	RoMa [14]	–	82.0	86.5	92.5	77.0
		✓	93.5	95.5	97.0	78.0
	RoMa	–	7.0	15.0	39.5	76.0
	(MatchAnything [15])	✓	96.5	<u>98.5</u>	<u>99.5</u>	76.9
Global	HSM-Net [18]	–	87.0	89.0	92.0	22.57 ³⁾
		✓	87.5	88.0	90.5	23.10 ³⁾
	AnyLoc [9]	–	<u>98.5</u>	99.5	100	0.586
	Revisit Anything [22]	–	99.0	99.5	100	<u>1.688</u>

²⁾ Geometric verification with MAGSAC++ [36]

³⁾ Tested on our re-implementation because the original code is not publicly available. The original paper reports 18.38 s per query.

Moreover, global descriptors aggregated from patch-level features could be sensitive to low overlap between a query and a reference tile. We therefore designed controlled experiments to isolate the effects of coverage, rotation, and aspect ratio variations on global-feature VPR performance.

A. Experiment 1: Local Feature vs. Global Feature for VPR

We evaluated representative local and global feature approaches on the VPAIR dataset [35]. We used 200 query images, 200 reference images, and 200 distractors. We reported Recall@ K for $K \in \{1, 2, 5\}$ with a distance threshold of 100 m, along with processing time per query. For local feature methods, we tested putative matching and geometric verification. For global feature methods, we evaluated AnyLoc [9] and Revisit Anything [22] as efficient retrieval baselines. All experiments were conducted on an NVIDIA GeForce RTX 4090 GPU.

To quantify local feature matching, we computed the putative matching score SC_c as

$$SC_c = \sqrt{\left(\frac{n_p}{n_{th}}\right)^2 + \left(\frac{n_M}{n_{th}}\right)^2} \quad (1)$$

where n_p denotes the count of feature points with a matching confidence > 0.6 , and $n_{th} = 500$ is the minimum threshold. n_M represents the total number of matched feature points. For geometric verification, we used MAGSAC++ [36] with a homography model (2 pixels reprojection error, 10,000 iterations, 0.995 confidence).

Table I summarizes the results. The results demonstrated that global feature approaches achieved both robust recall and high processing efficiency. In particular, RoMa (MatchAnything) [15] showed poor performance in putative matching, which may be related to overconfident matches in our setup. These findings suggest that global retrieval provides a robust backbone for online wide-area search, justifying our strategy of leveraging it to ensure scalability.

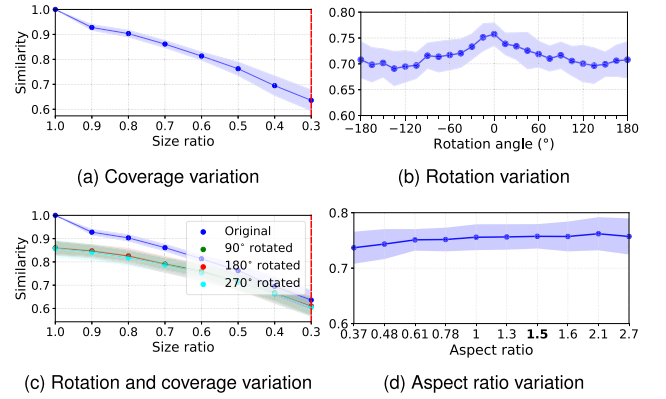


Fig. 1. Effect of geometric variations on global feature similarity: The experiments were conducted using 11 images with distinct visual appearances. The $\bullet$ symbol denotes the mean, and the shaded region represents the standard deviation. (a) As coverage decreases, similarity scores drop significantly, indicating that coverage variation is a major challenge for tile-map-based VPR. (b) Similarity evaluation where the reference image is center cropped at 50% coverage, and the query image is rotated from -180° to 180° in 15° increments; similarity sharply declines for rotations exceeding 90° . (c) Similarity evaluation under varying coverage (30% to 100% in 10% increments) at discrete rotation angles (0° , 90° , 180° , and 270°); similarity degradation due to coverage loss is more severe than that caused by rotation. (d) The results indicate little performance degradation across a wide range of aspect ratios.

B. Experiment 2: Coverage Variation for Global Feature

Tile maps are stored as fixed field-of-view (FoV) tiles, while UAV images can overlap with only a subset of a tile. This causes large coverage variation, which can be more severe than viewpoint changes in standard VPR benchmarks. To isolate this factor, we generated coverage-controlled queries by cropping the center region of an image while preserving the aspect ratio. The coverage ratio was swept from 30% to 100% in steps of 10%. We then evaluated descriptor similarity between the original image and the cropped image.

As shown in Fig. 1a, we observed that global-descriptor performance degraded significantly as coverage decreased. This result highlighted coverage variation as a key failure mode for tile map-based localization and motivated the use of multi-zoom, overlapped tiles to increase the chance that at least one reference tile provides sufficient overlap with the query.

C. Experiment 3: Rotation Variation for Global Feature

Next, we examined how global descriptors behave under rotation, which can occur when UAV heading differs from the orientation of satellite tiles. We conducted two tests. First, we cropped the reference to 50% coverage and rotated the query image from -180° to 180° in 15° steps and evaluated retrieval against the original reference. Second, we combined coverage variation (30% to 100%) with rotations of $\{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$ to compare the relative impact of rotation and coverage.

As shown in Fig. 1b and 1c, the results showed clear degradation under large rotations, with a notable drop beyond 90° . More importantly, the combined test indicated that coverage variation typically caused a larger performance loss than rotation alone. These findings justified two complementary design choices. We

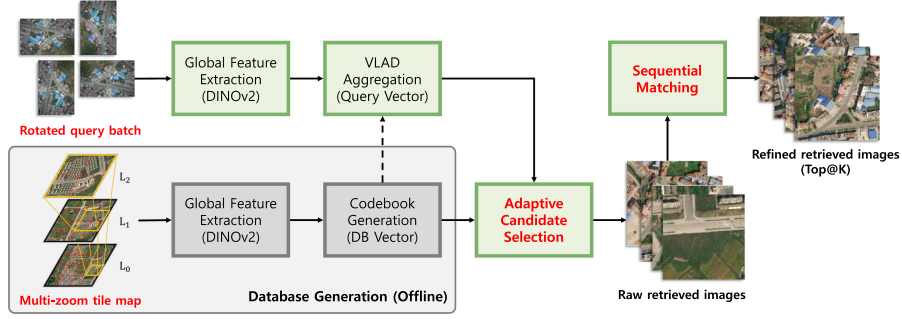


Fig. 2. Overview of TileLoc: TileLoc builds a multi-zoom, overlapped tile database, uses a rotated query batch, selects candidates by adaptive cumulative similarity mass, and aggregates sequential log-odds evidence with local-search mode switching. The four pipeline modules are described in Sections IV-B–IV-E.

explicitly handled rotation by using rotated query batches. We treated overlap (coverage) as the primary issue and prioritized multi-zoom, overlapped tile references, which are critical for stable retrieval under tile-map conditions.

D. Experiment 4: Aspect Ratio Variation for Global Feature

Finally, we tested whether aspect ratio mismatch is a major source of inaccuracy for global descriptors. The query aspect ratio was sampled around 1.0 with an approximate scaling factor of 1.28, while additionally including the original aspect ratio of 3:2. Specifically, $\frac{\text{width}}{\text{height}} \in \{0.37, 0.48, 0.61, 0.78, 1.0, 1.3, 1.5, 1.6, 2.1, 2.7\}$.

Unlike coverage and rotation, aspect-ratio changes caused little degradation in our tests. This suggests that aspect ratio is not a dominant factor for our pipeline, compared to coverage variation and related scale mismatch. Therefore, in the rest of this work, we focus on mitigating coverage-driven mismatch and stabilizing retrieval over time using sequential evidence aggregation.

Summary: Preliminary results defined the key requirements for TileLoc and motivated a global retrieval backbone that explicitly handles geometric mismatches. Experiment III-A showed that global feature approaches achieved higher recall with efficient computation in wide-area search. Experiments III-B and III-C identified that coverage loss caused the largest drop, and rotation further reduced similarity. To address these issues, TileLoc uses multi-zoom, overlapped tile maps and rotated query batches. Finally, we stabilize predictions with temporal evidence aggregation by accumulating retrieval scores as votes in tile-quantized space.

IV. TILELOC: SEQUENTIAL VPR WITH TILE MAPS

A. Problem Formulation

Our goal is to estimate a UAV’s global position as its *tile index* using onboard camera images and a database of web tile maps. Given a query image stream $\{q_n\}_{n=1}^N$ captured at unknown locations, TileLoc retrieves candidate tiles and aggregates evidence over time to produce a stable global estimate. As shown in the preliminary experiments, VPR under tile-map conditions is particularly sensitive to limited overlap (coverage mismatch)

and large in-plane rotations. TileLoc mitigates these issues while remaining efficient for wide-area search.

Overall Procedure: Fig. 2 summarizes the pipeline. TileLoc consists of four pipeline modules. First, we build a multi-zoom, overlapped tile database to increase the chance of spatial overlap between the query and tile images. Second, we use a rotated query batch to handle large in-plane rotations. Third, adaptive cumulative-mass candidate selection chooses a variable number of candidates according to retrieval ambiguity. Fourth, sequential log-odds evidence aggregation with local-search mode switching accumulates per-frame retrieval evidence in tile space. For tiles and queries, we use AnyLoc as the descriptor backbone: each image is encoded with DINOv2 features and aggregated into a VLAD descriptor with $K = 32$ clusters. Tile descriptors are precomputed offline and indexed with FAISS [37] for fast retrieval. Throughout this section, similarity refers to cosine similarity between VLAD descriptors, denoted by $s \in [0, 1]$.

B. Multi-Zoom Overlapped Tile Database

Web tile maps are provided as fixed FoV tiles, whereas UAV images may overlap with only a subset of a tile. To mitigate this coverage mismatch, we construct overlapped tiles at multiple zoom levels. Starting from standard 256×256 tiles at a reference zoom $z_{\max}$, we build 512×512 tiles by merging adjacent 2×2 tiles with a sliding window and stride of one tile. This yields 50% overlap between neighbors. We apply the same overlap strategy across zooms. For lower zooms, we first merge coarser tiles and then apply the sliding merge. For higher zooms, we crop 256×256 regions with 50% overlap and upscale them to 512×512 .

We choose $z_{\max}$ from camera parameters and a coarse altitude prior. Let f be focal length in pixels, (W, H) image resolution, φ latitude, and a_{prior} a prior altitude. We compute a conservative FoV using the shorter side as

$$\text{fov}_{\min} = 2 \arctan\left(\frac{\min(W, H)}{2f}\right), \quad (2)$$

and approximate the ground coverage (meters) under a pinhole model as

$$g_{\text{cov}} = 2a_{\text{prior}} \tan\left(\frac{\text{fov}_{\min}}{2}\right). \quad (3)$$

Under the web Mercator convention, the ground resolution at latitude φ is proportional to $\cos \varphi$ and scales as 2^{-z} . Using this relation, we estimate the zoom that best matches the query footprint as

$$z^* = \log_2 \left(\frac{\cos(\varphi) 2\pi R}{g_{\text{cov}}} \cdot \frac{\text{tile_size}}{256} \right),$$

$$z_{\text{max}} = \lfloor z^* + 0.5 \rfloor, \quad (4)$$

where R is the radius of the Earth and `tile_size` is the pixel size of our merged tile representation (512 in our implementation). We then build the multi-zoom, overlapped database around z_{max} to cover plausible zoom mismatch while keeping the database size manageable.

C. Rotated Query Batch

Large in-plane rotations can degrade global-feature based VPR. To improve robustness, for each query frame q_n we generate rotated queries as

$$q_n^\theta = \text{Rotate}(q_n, \theta), \quad \theta \in \Theta,$$

$$\Theta = \{0^\circ, 90^\circ, 180^\circ, 270^\circ\}. \quad (5)$$

Each rotated query is encoded into a descriptor, and compared against tile descriptors using cosine similarity. To avoid duplicate tiles while retaining the best rotation, we apply tile-wise max pooling over rotations as

$$s_n(i) = \max_{\theta \in \Theta} \text{sim}(q_n^\theta, d_i), \quad (6)$$

where d_i is the descriptor of tile i and $\text{sim}(\cdot, \cdot) \in [0, 1]$.

D. Adaptive Cumulative-Mass Candidate Selection

Instead of using a fixed top- k for every query, we dynamically adapt the number of candidates based on the concentration of similarity scores. Let $I_n^\downarrow = (i_1, i_2, \dots)$ be the sorted database tile indices such that $s_n(i_1) \geq s_n(i_2) \geq \dots$. Here, $s_n(i_j)$ is the same tile-wise similarity score from Equation (6), evaluated at the j -th ranked tile index. We then choose the smallest number of candidates k_n whose cumulative similarity mass reaches a threshold τ :

$$k_n = \min \left\{ k : \sum_{j=1}^k s_n(i_j) \geq \tau \right\}, \quad \tau = 2.0. \quad (7)$$

The candidate set for frame n is $\mathcal{C}_n = \{i_1, i_2, \dots, i_{k_n}\}$. The value k_n is not a fixed hyperparameter. It varies for each frame according to the similarity distribution. When the similarity distribution is sharply peaked, k_n remains small and focused; when the distribution is ambiguous, k_n increases to broaden the candidate set.

E. Sequential Matching with Log-Odds Accumulation

Single-frame retrieval can become ambiguous due to the presence of repetitive and visually similar tiles. Since tile maps are indexed by space rather than time, we accumulate evidence directly in tile space using a vote map $V_n(i)$.

Positive log-odds vote: For each candidate tile $i \in \mathcal{C}_n$, we map the similarity score $s_n(i)$ to a bounded probability through

Algorithm 1: Mode switching with hysteresis.

Require: Best tile i_n^* , Local radius r
 Previous mode $m_{n-1} \in \{\text{GLOBAL}, \text{LOCAL}\}$,
 Thresholds T_{enter} and T_{exit} ($T_{\text{enter}} > T_{\text{exit}}$)

- 1: $V_n^* \leftarrow V_n(i_n^*) \triangleright$ The peak vote score
- 2: $m_n \leftarrow m_{n-1}$
- 3: **if** $m_{n-1} = \text{LOCAL}$ **then**
- 4: Restrict retrieval to the $(2r + 1) \times (2r + 1)$ window centered at i_n^*
- 5: **if** $V_n^* \leq T_{\text{exit}}$ **then**
- 6: $m_n \leftarrow \text{GLOBAL} \triangleright$ Switch back to global search
- 7: **end if**
- 8: **else if** $m_{n-1} = \text{GLOBAL}$ **then**
- 9: Retrieve from the full database
- 10: **if** $V_n^* \geq T_{\text{enter}}$ **then**
- 11: $m_n \leftarrow \text{LOCAL} \triangleright$ Enter local search
- 12: **end if**
- 13: **end if**

a scaled sigmoid as

$$p(s) = p_{\min} + (p_{\max} - p_{\min}) \sigma(\alpha(s - s_m)),$$

$$\sigma(x) = \frac{1}{1 + e^{-x}}, \quad (8)$$

where s_m denotes the similarity value at which the sigmoid is centered, and α controls its slope. We then compute the corresponding log-odds as

$$\ell(s) = \log \frac{p(s)}{1 - p(s)}. \quad (9)$$

We accumulate votes in the log-odds domain as $V_n(i)$. To ensure non-negative evidence, we clamp the minimum probability to $p_{\min} = 0.5$, which yields $\ell(s) \geq 0$. At each step, competing hypotheses are suppressed through a temporal decay, and the votes are updated as

$$V_n(i) = \max(0, V_{n-1}(i) - \lambda + 1[i \in \mathcal{C}_n] \ell(s_n(i))). \quad (10)$$

Here, $1[i \in \mathcal{C}_n]$ is an indicator function that equals 1 if tile i belongs to the candidate set $\mathcal{C}_n$, and 0 otherwise. The index i in $V_n(i)$ and $V_{n-1}(i)$ denotes the same fixed database tile index, not a frame-specific candidate label. λ controls the log-odds decay and is set to $\lambda \approx 0.405$ in all experiments, corresponding to $p_{\text{decay}} = 0.6$. The per-frame prediction is given by $i_n^* = \arg \max_i V_n(i)$, and the peak vote score is $V_n^* = V_n(i_n^*)$.

Mode switching with hysteresis: TileLoc uses two retrieval modes: *global search* and *local search*. Global search retrieves from the full database, and local search restricts retrieval to a square neighborhood around the current best tile i_n^* (a fixed $(2r + 1) \times (2r + 1)$ window in tile coordinates). TileLoc switches modes using hysteresis on the peak vote score V_n^* to prevent rapid oscillation.

Because Equation (10) always decays the evidence of the vote map, the system returns to global search when evidence fades. Hysteresis prevents frequent mode flipping near the threshold.

Implementation details: We use DINOv2 ViT-G as the feature extractor and aggregate its features with VLAD, following the descriptor pipeline in the preliminary experiments. We use

TABLE II
PERFORMANCE EVALUATION ON THE UAV-VISLOC AND VPAIR DATASETS

Method	Distance Threshold	UAV-VisLoc															VPAIR		
		Sequence 1			Sequence 2			Sequence 3			Sequence 4			Sequence 9			300 samples		
		top-1	top-2	top-5	top-1	top-2	top-5	top-1	top-2	top-5	top-1	top-2	top-5	top-1	top-2	top-5	top-1	top-2	top-5
Baseline (AnyLoc [9])	100 m	10.89	14.44	18.48	11.86	15.03	18.02	11.07	14.84	22.01	10.43	13.82	18.97	9.4	14.62	22.58	16.33	24.33	36.67
	250 m	33.9	44.43	62.18	38.10	47.99	60.13	26.43	35.55	51.30	31.17	40.92	55.42	33.94	47.91	63.58	32.00	45.00	58.33
Baseline + O	100 m	+0.37	+1.10	+5.14	+0.18	+1.50	+5.60	+1.95	+3.26	+1.56	+6.51	+8.13	+7.86	+7.70	+10.05	+10.06	+3.00	+4.34	+3.33
	250 m	-1.71	-2.94	-3.67	+1.58	+1.59	+1.49	+11.72	+12.11	+10.68	+7.58	+9.35	+4.88	+15.15	+14.23	+14.36	+1.00	+0.67	+1.67
Baseline + O, Z	100 m	+29.26	+35.50	+45.29	+18.77	+25.96	+35.01	+18.23	+25.78	+30.46	+12.20	+17.35	+25.20	+7.70	+10.05	+10.06	+29.00	+32.00	+30.33
	250 m	+25.95	+22.16	+15.05	+11.76	+8.97	+5.32	+22.01	+20.96	+16.41	+12.87	+11.52	+7.05	+15.15	+14.23	+14.36	+22.00	+18.33	+16.00
Baseline + O, Z, R	100 m	+41.50	+49.57	+55.57	+29.04	+38.56	+44.73	+34.24	+41.67	+48.95	+21.41	+30.22	+39.70	+9.92	+11.62	+12.28	+31.34	+35.34	+32.00
	250 m	+36.72	+31.58	+20.56	+22.12	+15.88	+9.99	+35.81	+35.67	+29.30	+25.74	+24.66	+19.65	+19.06	+18.41	+17.23	+24.67	+21.33	+18.00
Baseline + O, Z, R, L	100 m	+41.62	+49.57	+55.82	+28.94	+38.56	+44.63	+33.98	+41.15	+49.21	+21.14	+30.35	+39.70	+10.57	+14.36	+14.10	+31.34	+36.67	+34.33
	250 m	+36.60	+31.58	+20.56	+22.12	+15.88	+10.18	+36.07	+35.67	+29.43	+25.88	+25.07	+19.78	+25.59	+25.33	+20.88	+26.67	+23.67	+21.00

Baseline rows are absolute recall (%) and all other rows are gains over AnyLoc in percentage points (%p). **O**: overlapped database, **Z**: multi-zoom database, **R**: rotated query batch, and **L**: local search with sequential matching mode switching.

cosine similarity for retrieval. Unless stated otherwise, we use $\Theta = \{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$, $\tau = 2.0$, and $K = 32$ VLAD clusters. The parameters T_{enter} , T_{exit} , and r are determined using grid search on UAV-VisLoc [38].

V. EXPERIMENTS

We evaluated TileLoc for aerial VPR using tile maps. The ablation study uses four incremental variants: **O** (overlapped database), **Z** (multi-zoom database), **R** (rotated query batch), and **L** (local search with sequential matching mode switching). Adaptive cumulative-mass candidate selection is part of the retrieval pipeline and is not separately ablated. We also analyzed precision-recall behavior and runtime.

A. Experimental Setup

We conducted experiments on two UAV localization datasets, UAV-VisLoc [38] and VPAIR [35]. We evaluated sequences 1, 2, 3, 4, and 9, which include long-range motion and significant appearance variability. Flight altitude ranged approximately from 400 m to 800 m. As reference imagery, we built a tile map database from ESRI satellite imagery. We computed recall by comparing the geodetic position of the retrieved reference tile with the ground-truth position of each query.

To isolate the impact of each ablation factor, we used AnyLoc [9] as the baseline. The baseline used standard non-overlapped tiles (256×256) at a single zoom. It performed single-frame retrieval without rotation handling or temporal aggregation. We then added ablation factors one by one and measured the gain.

We measured top-1, top-2, and top-5 recall under distance thresholds of 100 m and 250 m. A prediction is correct if its distance error is within the threshold. These thresholds evaluate robustness under different error tolerances for tile-based localization.

For UAV-VisLoc, we evaluated sequences 1, 2, 3, 4, and 9. We excluded sequences that were too small or dominated by repetitive natural scenes (e.g., rivers, mountains, and deserts), where location was difficult to pinpoint even with recorded UAV images. For VPAIR, we reported a 300-sample subset due to the cost of building and indexing the full tile-map database.

B. Performance Evaluation

Table II summarizes the ablation results over five sequences. We use the following notation: **O** (overlapped database at the base zoom), **Z** (multi-zoom extension with zoom estimation), **R** (rotated query batch), and **L** (sequential matching with mode switching for local search).

Effect of overlap and multi-zoom: Adding overlapped tiles (**O**) improved recall across sequences, particularly at larger distance thresholds. This indicates that overlap mitigates coverage mismatch by increasing the likelihood that at least one reference tile shares sufficient visible area with the query. The largest gain from **O** was observed in sequence 9 at 250 m, where Recall@1 improved by 15.15%p (from 33.94% to 49.09%). This suggests that overlap is especially beneficial under coarse localization, where small spatial misalignments are more significant.

Adding multi-zoom retrieval with zoom estimation (**Z**) produced a substantial improvement on several sequences. In particular, **Z** produced the largest single-factor gain in our ablation on sequence 1 at 100 m, where Recall@5 increased by 40.15%p (from 23.62% to 63.77%). This supports the importance of selecting a zoom level consistent with the query footprint when the reference map is provided at discrete zooms, and suggests that coverage/scale mismatch is a dominant error source in VPR with tile maps.

Effect of rotated query batch: UAV heading can change frequently. Enabling rotation hypotheses (**R**) consistently improved recall, and the largest gain appeared on sequence 3 at 100 m, where Recall@5 increased by 18.49%p (from 52.47% to 70.96%). This matches the preliminary finding that global descriptors are sensitive to large in-plane rotations, and shows that explicit rotation search is effective.

Effect of sequential matching and local search: The full system with sequential matching and mode switching (**L**) achieved the best overall performance, reaching 84.46% Recall@5 at 250 m on sequence 9. While the gain over the **O+Z+R** variant is modest in some settings, **L** primarily targets stability under visual aliasing and ambiguous terrain by suppressing inconsistent single-frame outliers. On sequence 9 at 250 m, **L** improved Recall@1 by 6.53%p (from 53.00% to 59.53%) and Recall@2 by 6.92%p (from 66.32% to 73.24%). These gains indicate that **L** mainly contributes through temporal stabilization. In addition, once the confidence is sufficiently high, local search

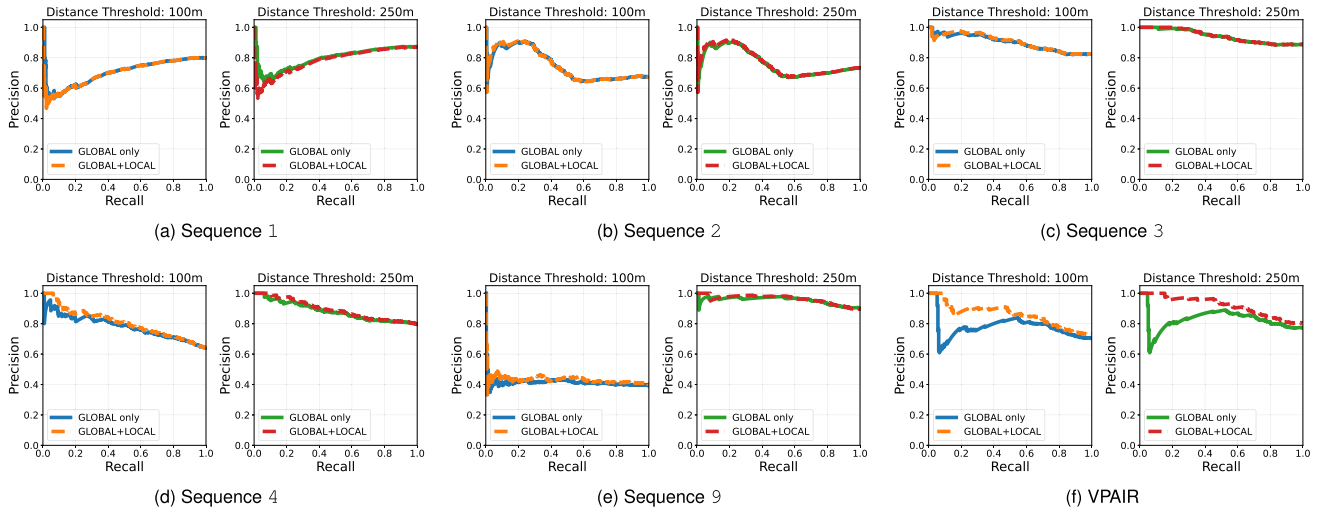


Fig. 3. Precision-recall curves for different sequences at distance thresholds of 100 m and 250 m: Precision and recall are computed by varying the confidence threshold on the peak vote score. A prediction is considered a true positive if the distance error to the ground truth is within the specified threshold: (a) Sequence 1 (b) Sequence 2 (c) Sequence 3 (d) Sequence 4 (e) Sequence 9 (f) VPAIR.

focuses retrieval on a small neighborhood, which can reduce unnecessary comparisons in subsequent frames.

Notes on thresholds: TileLoc uses a small set of decision thresholds for mode switching (e.g., T_{enter} and T_{exit}) and the local window radius r . In practice, the operating point of these thresholds depends on the score distribution, which can vary with terrain composition and sequence characteristics. We selected them by grid search and reported results using the selected operating point.

C. Precision and Recall Analysis

Fig. 3 shows the precision-recall (PR) curves for each sequence at distance thresholds of 100 m and 250 m. We varied the confidence threshold using the peak vote score V_n^* . A prediction is considered a true positive if the distance error is within the threshold. Overall, TileLoc achieved higher precision at comparable recall than global-only retrieval, indicating that sequential aggregation improves reliability. This matches the ablation results and supports the role of sequential matching and mode switching. A common failure mode occurred in vast, featureless regions such as wide rivers in sequence 1. Ambiguous tiles led to repeated false matches, and local refinement sometimes converged to the wrong region.

D. Runtime Analysis

To consider UAV Size, Weight, and Power (SWaP) constraints, we include runtime evaluation on both a desktop GPU (NVIDIA GeForce RTX 4090; 450 W TDP) and an edge platform (NVIDIA Jetson AGX Orin; 60 W TDP) using the UAV-VisLoc dataset. Table III reports latency and recall across DINOv2 backbones (ViT-G, ViT-L, ViT-B, and ViT-S) on both platforms. The results show that DINOv2 feature extraction is the dominant bottleneck, while retrieval and sequential matching add only minor overhead. With ViT-L, TileLoc runs in 193.0 ms on the RTX 4090 and 850.7 ms on the AGX Orin, with a

TABLE III
RUNTIME BREAKDOWN AND RECALL@250 M ACROSS BACKBONES

DINOv2 Backbone	Computing Platform	Runtime Breakdown [ms]				Recall@250 m		
		Feature Extraction	Retrieval	Sequential Matching	Total	top-1	top-2	top-5
ViT-G (1.1B)	RTX 4090	383.6	1.0	0.3	384.9	63.2	73.9	84.0
	AGX Orin	1939.2	16.1	7.8	1963.1			
ViT-L (0.3B)	RTX 4090	191.9	0.8	0.3	193.0	44.8	58.5	71.7
	AGX Orin	834.9	11.8	4.0	850.7			
ViT-B (86M)	RTX 4090	116.8	2.4	2.3	121.5	37.2	49.5	64.2
	AGX Orin	388.7	9.6	3.2	401.5			
ViT-S (21M)	RTX 4090	82.1	1.6	1.5	85.2	18.3	25.5	35.9
	AGX Orin	240.0	3.3	3.3	246.6			

Recall@5 decrease from 84.0% to 71.7% at the 250 m threshold. These runtimes are appropriate for low-rate global relocalization, drift correction, or navigation backup, rather than low-level flight stabilization.

VI. CONCLUSION

This letter presents TileLoc, a visual place recognition framework that utilizes lightweight tile maps for UAV navigation. TileLoc tackles geometric and appearance mismatches through a multi-zoom, overlapped tile database, rotated query batches, adaptive cumulative-mass candidate selection, and sequential log-odds evidence aggregation. Experiments on two public datasets demonstrate that these modules improve recall over the baseline, and the runtime analysis indicates that TileLoc can serve low-rate global relocalization, drift correction, and navigation backup tasks on edge hardware.

Limitations: Despite these advancements, TileLoc has certain limitations. First, its applicability is constrained in low-texture environments such as dense forests or rivers, where distinctive visual features are scarce. Moreover, when the UAV is close to the ground (e.g., during landing), TileLoc may fail due to a lack

of visual correspondence with tile images. Second, the database must be updated regularly to reflect structural changes over time (e.g., new roads and building complexes). While TileLoc can tolerate a certain degree of appearance discrepancy thanks to the robust representation of high-level global features, it cannot cope with sudden and extensive changes (e.g., large-scale disaster situations). In the event of such abrupt and large-scale changes, the system must utilize up-to-date imagery acquired from commercial satellite vendors rather than relying on publicly available web tiles.

Future work: Future work will address these limitations by incorporating visual-inertial odometry to provide motion constraints to handle low-texture or ambiguous terrains. Ultimately, we aim to develop a comprehensive UAV localization pipeline that seamlessly integrates TileLoc with onboard odometry, providing robust and accurate positioning across diverse operational scenarios. Additionally, TileLoc itself can be further enhanced. Since our framework is designed by integrating four additional components onto the baseline VPR method, AnyLoc, we plan to upgrade this underlying baseline to more recent state-of-the-art methods in future.

REFERENCES

- [1] P. J. Teunissen and O. Montenbruck, *Springer Handbook of Global Navigation Satellite Systems*, vol. 10. Berlin, Germany: Springer, 2017.
- [2] O. Y. Al-Jarrah, A. S. Shatnawi, M. M. Shurman, O. A. Ramadan, and S. Muhaidat, "Exploring deep learning-based visual localization techniques for UAVs in GPS-denied environments," *IEEE Access*, vol. 12, pp. 113049–113071, 2024.
- [3] Z. Lu, F. Liu, and X. Lin, "Vision-based localization methods under GPS-denied conditions," 2022, *arXiv:2211.11988*.
- [4] M. Faessler, F. Fontana, C. Forster, E. Mueggler, M. Pizzoli, and D. Scaramuzza, "Autonomous, vision-based flight and live dense 3D mapping with a quadrotor micro aerial vehicle," *J. Field Robot.*, vol. 33, no. 4, pp. 1–20, 2016.
- [5] S. B. Laina et al., "Scalable outdoors autonomous drone flight with visual-inertial SLAM and densesubmaps built without LiDAR," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2025.
- [6] A. Steenbeek and F. Nex, "CNN-based dense monocular visual SLAM for real-time UAV exploration in emergency conditions," *Drones*, vol. 6, no. 3, 2022, Art. no. 79.
- [7] I. Cvišić, J. Česić, I. Marković, and I. Petrović, "SOFT-SLAM: Computationally efficient stereo visual simultaneous localization and mapping for autonomous unmanned aerial vehicles," *J. Field Robot.*, vol. 35, no. 4, pp. 578–595, Jun. 2018.
- [8] A. Gupta and X. Fernando, "Simultaneous localization and mapping (SLAM) and data fusion in unmanned aerial vehicles: Recent advances and challenges," *Drones*, vol. 6, no. 4, 2022, Art. no. 85.
- [9] N. Keetha, et al., "AnyLoc: Towards universal visual place recognition," *IEEE Robot. Automat. Lett.*, vol. 9, no. 2, pp. 1286–1293, Feb. 2024.
- [10] I. Moskalenko, A. Kornilova, and G. Ferrer, "Visual place recognition for aerial imagery: A survey," *Robot. Auton. Syst.*, vol. 183, 2025, Art. no. 104837.
- [11] D. DeTone, T. Malisiewicz, and A. Rabinovich, "SuperPoint: Self-supervised interest point detection and description," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops*, 2018, pp. 337–33712.
- [12] P. Lindenberger, P.-E. Sarlin, and M. Pollefeys, "LightGlue: Local feature matching at light speed," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 17581–17592.
- [13] Y. Wang, X. He, S. Peng, D. Tan, and X. Zhou, "Efficient LoFTR: Semi-dense local feature matching with sparse-like speed," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 21666–21675.
- [14] J. Edstedt, Q. Sun, G. Bökman, M. Wadenbäck, and M. Felsberg, "RoMa: Robust dense feature matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 19790–19800.
- [15] X. He, et al., "MatchAnything: Universal cross-modality image matching with large-scale pre-training," 2025, *arXiv:2501.07556*.
- [16] A. Şanlı, H. T. Bağcı, and E. Koyuncu, "Particle filter-based localization using visual feature synchronization in GNSS-denied navigation," in *Proc. Integr. Commun., Navigation Surveill. Conf.*, 2025, pp. 1–8.
- [17] Y. Di, et al., "A UAV location method based on aerial-satellite image matching and factor graph optimization in GNSS-denied area," *Meas. Sci. Technol.*, vol. 36, no. 11, 2025, Art. no. 116302.
- [18] Z. Zhang, J. Chu, T. Song, J. Guo, R. Zhang, and J. Li, "GeoRVLF: A robust drone-satellite visual geo-localization framework for small UAV platforms," *IEEE Robot. Automat. Lett.*, vol. 10, no. 7, pp. 7380–7387, Jul. 2025.
- [19] H. Jégou, M. Douze, C. Schmid, and P. Pérez, "Aggregating local descriptors into a compact image representation," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, 2010, pp. 3304–3311.
- [20] R. Arandjelovic, P. Gronat, A. Torii, T. Pajdla, and J. Sivic, "NetVLAD: CNN architecture for weakly supervised place recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 5297–5307.
- [21] S. Chen, X. Wu, M. W. Mueller, and K. Sreenath, "Real-time geo-localization using satellite imagery and topography for unmanned aerial vehicles," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2021, pp. 2275–2281.
- [22] K. Garg, S. S. Puligilla, S. Kolathaya, M. Krishna, and S. Garg, "Revisit Anything: Visual place recognition via image segment retrieval," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 1–19.
- [23] P. Ai, et al., "MM-Geo: Multi-scale and multi-positive UAV-view geo-localization," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2025, pp. 9784–9791.
- [24] S. Schubert, P. Neubert, S. Garg, M. Milford, and T. Fischer, "Visual place recognition: A tutorial," *IEEE Robot. Automat. Mag.*, vol. 31, no. 3, pp. 139–153, Sep. 2024.
- [25] M.-A. Tomiä, M. Zaffar, B. Ferrarini, M. J. Milford, K. D. McDonald-Maier, and S. Ehsan, "Sequence-based filtering for visual route-based navigation: Analyzing the benefits, trade-offs and design choices," *IEEE Access*, vol. 10, pp. 81974–81987, 2022.
- [26] M. J. Milford and G. F. Wyeth, "SeqSLAM: Visual route-based navigation for sunny summer days and stormy winter nights," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2012, pp. 1643–1649.
- [27] P. Hansen and B. Browning, "Visual place recognition using HMM sequence matching," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2014, pp. 4549–4555.
- [28] S. Garg and M. Milford, "SeqNet: Learning descriptors for sequence-based hierarchical place recognition," *IEEE Robot. Automat. Lett.*, vol. 6, no. 3, pp. 4305–4312, Jul. 2021.
- [29] R. Mereu, G. Trivigno, G. Berton, C. Masone, and B. Caputo, "Learning sequential descriptors for sequence-based visual place recognition," *IEEE Robot. Automat. Lett.*, vol. 7, no. 4, pp. 10383–10390, Oct. 2022.
- [30] S. Garg, M. Vankadari, and M. Milford, "SeqMatchNet: Contrastive learning with sequence matching for place recognition & relocation," in *Proc. Conf. Robot Learn.*, 2021, pp. 429–443.
- [31] O. Vysotska and C. Stachniss, "Effective visual place recognition using multi-sequence maps," *IEEE Robot. Automat. Lett.*, vol. 4, no. 2, pp. 1730–1736, Apr. 2019.
- [32] M. Yang, J. Mao, X. He, L. Zhang, and X. Hu, "A sequence-based visual place recognition method for aerial mobile robots," *J. Phys.: Conf. Ser.*, vol. 1654, no. 1, 2020, Art. no. 012080.
- [33] M. Oquab et al., "DINOv2: Learning robust visual features without supervision," 2023, *arXiv:2304.07193*.
- [34] E. Pizzi, S. D. Roy, S. N. Ravindra, P. Goyal, and M. Douze, "A self-supervised descriptor for image copy detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 14532–14542.
- [35] M. Schleiss, F. Rouatbi, and D. Cremers, "VPAIR–Aerial visual place recognition and localization in large-scale outdoor environments," 2022, *arXiv:2205.11567*.
- [36] D. Barath, J. Noskova, M. Ivashechkin, and J. Matas, "MAGSAC, a fast, reliable and accurate robust estimator," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 1301–1309.
- [37] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," *IEEE Trans. Big Data*, vol. 7, no. 3, pp. 535–547, Jul. 2021.
- [38] W. Xu et al., "UAV-VisLoc: A large-scale dataset for UAV visual localization," 2024, *arXiv:2405.11936*.